Comprehensive Implementation Manual

20MAD481/482: Mini Project

Computational Mathematics for Engineers

B.Tech Computer Science and Engineering

(7th Semester)

Department of Mathematics

Saintgits College of Engineering (Autonomous)

Kottayam, Kerala

This document serves as the definitive reference guide encompassing the official syllabus mandates, strategic execution guidelines, curated project themes, and a sample week-by-week Work Breakdown Structure.

Contents

1. Syllabus Essentials & Evaluation Structure

1.1. Administrative Committees

The evaluation of the mini-project is strictly governed by two designated committees to ensure fairness and adherence to academic standards:

- **Internal Project Evaluation Committee (PEC):** Responsible for Continuous Internal Evaluation (CIE). Composed of the Head of Department (or a senior faculty member as Chairman), the Mini Project Coordinator, and the respective Project Supervisor.
- **End Semester Examination (ESE) Committee:** Responsible for the final defense. Composed of the HoD/Chairman, an **External Expert** (from Industry, Research, or Academia), and the Project Coordinator.

1.2. Complete Marking Scheme (150 Marks Total)

Students must score a minimum of **75 marks** overall to pass the mini-project. The evaluation is distributed as follows:

Mode	Evaluator	Evaluation Component	Marks
CIE	Project Supervisor	Project Scheduling (5), Literature Survey (5), Project Diary/Log (10), Completion/Demo (5)	25
	PEC	**Interim Evaluation:** Technical Content (7.5) & Presentation (7.5)	15
	PEC	**Final Evaluation:** Technical Content (7) & Presentation (8)	15
	PEC	**Mini Project Report:** Tech Content (6), Quality (6), Formats (4), References (4)	20
ESE	ESE Committee	Technical Content (Evaluated alongside external expert)	60
	ESE Committee	Presentation and Oral Defense	15
		Total Minimum Required to Pass:	**75**

1.3. Mandatory Report Structure

The final project report **must** include the following sections exactly as prescribed by the syllabus: 1. Executive Summary, 2. Introduction, 3. Methodology, 4. Results, 5.

Discussion, 6. Conclusion, and 7. Recommendations. Reports must be approved by the supervisor prior to the Final Review and certified by the HoD prior to the ESE.

2. Strategic Implementation Guidelines

Given that 7th-semester CSE students are balancing major projects and other minor courses, this computational mathematics project is structured to maximize their existing programming skills while satisfying rigorous mathematical evaluation criteria.

2.1. Strategic Directives for Supervisors

- **Logbook Discipline (10 Marks):** The CIE allocates 10 marks to the daily/weekly activity diary. To ensure authenticity, establish a strict 5–10 minute window during the weekly 4-hour project block to review and sign these diaries.
- **Live Code Verification (5 Marks):** For the final 5 marks under the Supervisor's purview ("Completion of the project"), students must execute their computational models live to prove that the mathematical objectives were met.
- **Template Enforcement:** To secure the 8 marks awarded by the PEC for formatting and referencing, mandate the use of a standardized LaTeX template. This handles complex mathematical typesetting seamlessly.

2.2. Directives for Students

- **Technology Stack:** Utilize familiar Python data science libraries (NumPy, Matplotlib, SciPy, PyTorch) to minimize coding friction and keep the focus on the underlying computational mathematics.
- **Interim Focus:** The Week 7 Interim Review strictly targets the mathematical formulation and preliminary algorithmic setup. Limit presentations to 5–7 slides.
- **Post-Review Modifications:** Students have exactly five working days following the Final PEC Evaluation (Week 13) to incorporate suggested modifications before the final HoD certification.

3. Curated Project Themes

The following themes are designed to provide a high level of experiential learning in Machine Learning and Data Analytics by forcing students to engage with the underlying computational mathematics (matrices, optimization, calculus) rather than using black-box libraries.

1. Financial Risk Analytics & Portfolio Optimization

- **Narrative:** Model risk vs. reward by calculating how financial assets interact, determining optimal wealth distribution to maximize return for a given risk level.
- **Math Focus:** Covariance/correlation matrices, expected value vectors, quadratic programming (Markowitz Efficient Frontier).
- **Dataset:** Stock Portfolio Optimization (Kaggle)

2. Chaotic System Modeling & Weather Time-Series Forecasting

- **Narrative:** Predict atmospheric conditions by modeling highly chaotic physical systems with significant micro-variations and non-linear patterns.
- **Math Focus:** Statistical normalization, auto-correlation functions, time-step gradient calculations for recurrent networks.
- **Dataset:** Weather Long-term Time Series (Kaggle)

3. Multi-Criteria Supply Chain Logistics Optimization

- **Narrative:** Solve complex operational routing puzzles, assigning shipping pathways while managing conflicting costs and capacity constraints.
- **Math Focus:** Deterministic objective functions, Linear/Integer Programming matrix formulations, constraint-satisfaction inequalities.
- **Dataset:** Supply Chain Logistics Problem (Kaggle)

4. Recommender Systems via Low-Rank Matrix Factorization

- **Narrative:** Build a collaborative filtering engine uncovering hidden patterns in user preferences by decomposing highly sparse preference matrices.
- **Math Focus:** Singular Value Decomposition (SVD), Alternating Least Squares (ALS) optimization, iterative matrix approximation errors.
- **Dataset:** Book Recommendation Dataset (Kaggle)

5. Fraud Detection in Imbalanced Data Analytics

- **Narrative:** Identify rare anomalies within millions of normal operations, requiring the handling of severe statistical skewness.

- **Math Focus:** Principal Component Analysis (PCA), Mahalanobis distance metric calculation, AUPRC gradient computation.
- **Dataset:** Credit Card Fraud Detection (Kaggle)

6. **Unsupervised Text Analytics & Statistical Topic Modeling**

- **Narrative:** Process massive text corpora to mathematically uncover latent themes without human labeling.
- **Math Focus:** TF-IDF sparse vector space construction, Dirichlet distributions, Gibbs Sampling for LDA.
- **Dataset:** TripAdvisor Hotel Reviews (Kaggle)

7. **Predictive Maintenance & Survival Analysis in Industrial IoT**

- **Narrative:** Analyze continuous stream telemetry from factory sensors to calculate the exact mathematical probability of machine degradation.
- **Math Focus:** Weibull distribution estimation, probability density function survival modeling, multivariate regression.
- **Dataset:** AI4I 2020 Predictive Maintenance (Kaggle)

8. **Customer Segmentation via High-Dimensional Clustering**

- **Narrative:** Discover distinct consumer purchasing archetypes within raw behavioral logs to execute precise interventions.
- **Math Focus:** Voronoi iteration optimization (K-Means), Expectation-Maximization (EM) for GMMs, Elbow Method derivatives.
- **Dataset:** E-Commerce Customer Behavior (Kaggle)

9. **Stochastic Modeling for Urban Traffic Flow Simulation**

- **Narrative:** Simulate vehicle arrival behaviors and bottlenecks to optimize traffic light timing parameters.
- **Math Focus:** Poisson processes, continuous-time Markov chains, steady-state probabilities in Queuing Models.
- **Dataset:** Metro Interstate Traffic Volume (Kaggle)

10. **High-Dimensional Gene Expression Analysis**

- **Narrative:** Map out thousands of cellular gene vectors simultaneously to categorize biological mutations.
- **Math Focus:** High-dimensional covariance matrix analysis, dominant Eigenvalues/vectors, non-linear dimensionality reduction.
- **Dataset:** Cancer Genomics Gene Expression (Kaggle)

11. Grid-Scale Energy Consumption Forecasting

- **Narrative:** Model regional infrastructure energy demands over several years to minimize grid power overproduction waste.
- **Math Focus:** Linear difference equations, Maximum Likelihood Estimation (MLE), Augmented Dickey-Fuller stationarity tests.
- **Dataset:** PJM Hourly Energy Consumption (Kaggle)

12. Markovian Decision Making for Algorithmic Trading Engines

- **Narrative:** Build a dynamic computational agent that learns optimal sequences of decisions based on changing financial environments.
- **Math Focus:** Markov Decision Processes (MDP), Bellman Optimality equations, iterative value-iteration convergence.
- **Dataset:** Crypto Historical Data (Kaggle)

4. Sample Work Breakdown Structure (WBS)

Below is a precise, 14-week execution map for the **Financial Risk Analytics & Portfolio Optimization** project. This WBS guarantees that every syllabus milestone and evaluation rubric is met on time.

Phase & Timeline	Task Description & Math Focus	Evaluation Milestone / Mark Capture
Phase 1 (Weeks 1–2) *Initiation*	Define a basket of 10-15 assets. Extract 5 years of historical price data. Set up Python/NumPy environments.	Group formation & Dataset lockdown.
Phase 2 (Week 3) *Methodology*	Formulate objective function mathematically: maximizing Sharpe ratio while minimizing portfolio variance. Submit WBS.	**Proposal Submission.** Supervisor awards up to 5 marks for Project Scheduling.
Phase 3 (Weeks 4–6) *Modeling*	Calculate logarithmic returns. Construct the Covariance Matrix and Expected Return Vector natively in Python.	**CIE Begins.** Supervisor weekly diary sign-offs secure 10 logbook marks.
Phase 4 (Week 7) *Interim Review*	Compile the mathematical framework and initial statistical models into a 5–7 slide presentation.	**Interim PEC Review (15 Marks):** Focus on technical framework, literature survey, and presentation skills.
Phase 5 (Weeks 8–11) *Optimization*	Implement quadratic programming to solve for the optimal weight vector. Compute and plot the Efficient Frontier.	**Live Code Verification:** Supervisor awards final 5 marks for Project Completion based on functional code.
Phase 6 (Weeks 12–13) *Final Internal*	Document mathematical proofs and interpret the Efficient Frontier. Draft report formatted in LaTeX.	**Final PEC Review (15 Marks)** and Draft Report Evaluation **(20 Marks)**.
Phase 7 (Week 14) *External Defense*	Incorporate PEC modifications (within 5 days). Secure HoD certification. Prepare final oral defense.	**End Semester Examination (75 Marks):** Evaluated by ESE Committee and External Expert.